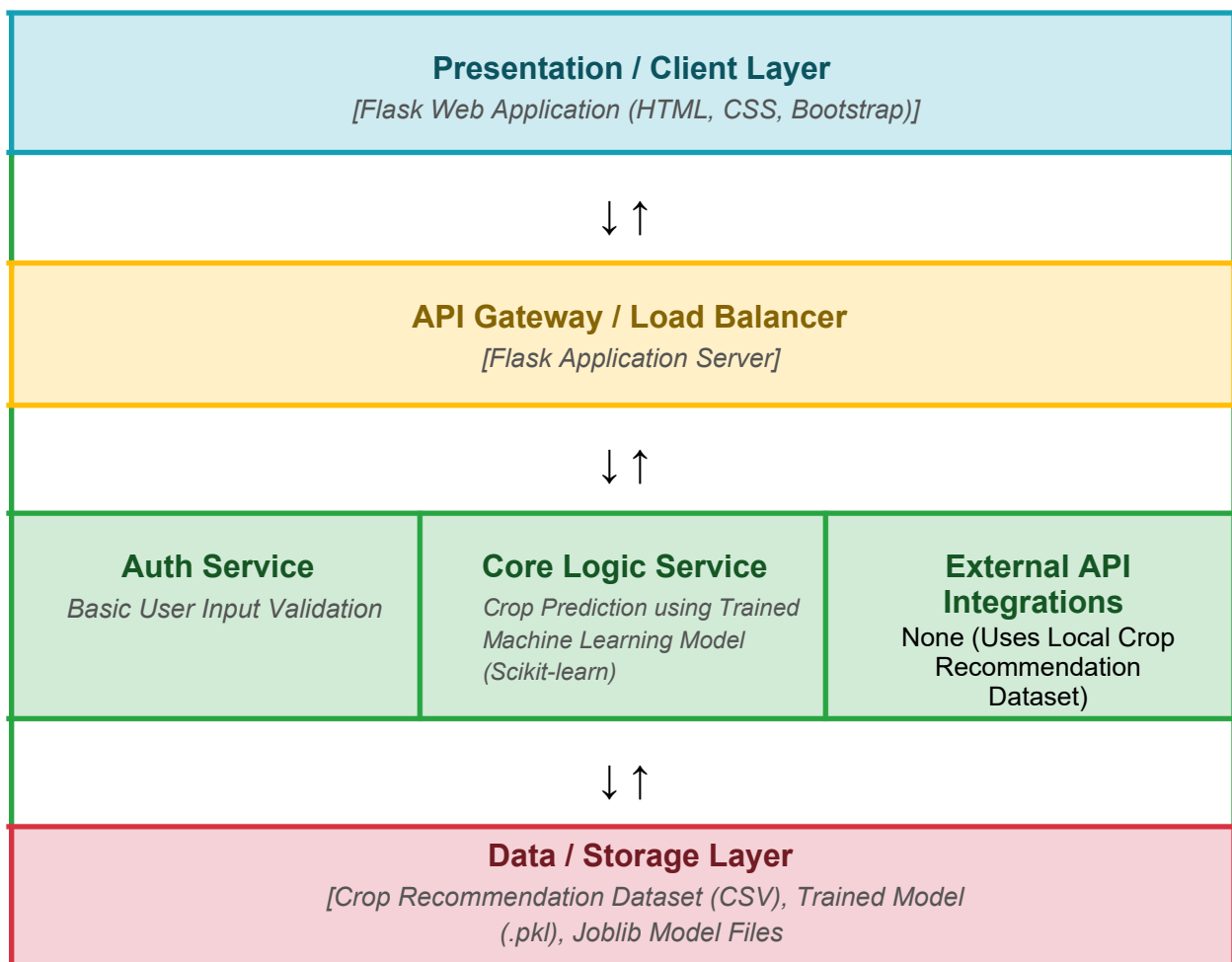


Solution Architecture

Date	12 July 2026 xxxxxx
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Collects soil and weather inputs from users and displays crop recommendations.</i>	<i>Flask, HTML, CSS, Bootstrap</i>
<i>[API Gateway]</i>	<i>Receives user requests and forwards them to the ML prediction service.</i>	<i>Flask</i>
<i>[Microservices]</i>	<i>Processes user input, loads the trained ML model, predicts the best crop, and returns the result.</i>	<i>Python, Scikit-learn, Pandas, NumPy, Joblib</i>
<i>[Database]</i>	<i>Stores the crop dataset and trained machine learning model used for prediction. CSV Dataset, Pickle (.pkl), Joblib</i>	<i>CSV Dataset, Pickle (.pkl), Joblib</i>